

Adaptive Attack on Static POKÉ Keys

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Abstract. Isogeny-based cryptosystems continue to show promise in post-quantum cryptography. In recent years, numerous constructions have been proposed, one of which is POKÉ, a compact and efficient public-key exchange system that uses higher-dimensional isogenies [2]. This paper leverages a well-known adaptive attack on SIDH [10], and demonstrated a similar attack on POKÉ, when given a key exchange oracle with the same assumptions as those posed by Galbraith et al. [10]. This attack relies on the user to employ long-term static keys which is against the intent of the designers of POKÉ. Indeed, this attack provides further evidence that POKÉ should not be used with a long-term static key.

1 Introduction

Isogeny-based cryptosystems have emerged as promising candidates in the post-quantum cryptographic landscape by offering compact key sizes and base their security assumptions on the hardness of finding isogenies between supersingular elliptic curves. These constructions rely on the presumed intractability of computing isogenies between supersingular elliptic curves, which is a problem believed to resist both classical and quantum attacks. The field gained momentum with the introduction of the Supersingular Isogeny Diffie–Hellman (SIDH) [12] key exchange protocol in 2011, followed by its evolution into the Supersingular Isogeny Key Encapsulation (SIKE) scheme. SIKE was notably selected as an alternate candidate in NIST’s Post-Quantum Cryptography Standardisation process but was broken in 2022, highlighting both the potential and the need for a thorough understanding of the security properties of isogeny-based cryptosystems.

During the past decade, the cryptographic community has scrutinized isogeny-based schemes from various angles, leading to a range of attacks that exploit mathematical or structural vulnerabilities. The most prominent example is breaking SIDH in a series of papers [4, 16, 19] using torsion point information, effectively removing SIKE from NIST’s standardization efforts. Since then, there has been a flurry of results to provide countermeasures to the breaks [1, 7], or to propose new schemes to avoid revealing torsion point images that doomed SIDH [2, 3, 18].

Before the catastrophic breakdown of SIDH, cryptanalysis work focused on auxiliary points. In 2016, Galbraith et al. [10] published an adaptive attack on SIDH when a party uses a long-term static private key. In the attack, the attacker sends carefully chosen auxiliary points to the victim and gains a single bit of information about the key and is able to recover the entire secret in fewer than $\log n$ interactions, where n is the secret to be recovered. Since then, there have been numerous adaptive attacks on various isogeny-based cryptosystems [5, 14, 17] that are based on the same principles as sending carefully crafted points to the victim and gaining information from the process.

In this paper, we propose yet another adaptive attack on an isogeny-based cryptosystem. The cryptosystem in question is POKÉ, a novel compact and efficient public-key exchange using higher-dimensional isogenies. POKÉ builds upon algorithms from cryptosystems such as SIDH [12] and QFESTA [18], using said dimensional isogenies to construct an SIDH-Square that mitigates conventional attacks on SIDH. A detailed and brief analysis, as well as the concepts used in POKÉ, will be covered in Sections 2 and 3.

In particular, this paper proposes a hypothetical scenario in which a party in the key exchange uses a constant private key and other parties perform key exchanges with the said member.

Even though key exchanges and shared secrets will differ between individuals, we demonstrate an adaptive attack that can be conducted to leak the static secret value. We will later show how this partial leak of

a private key can lead to the decryption of victim’s messages to other parties, thereby necessitating the use of Fujisaki–Okamoto transform [8, 11] or similar to ensure IND-CCA Security.

2 Preliminaries

For a more thorough and complete introduction to POKÉ, we refer the reader to its introductory paper [2]. For an in-depth overview of elliptic curves and isogenies, we refer the reader to [21].

We begin by covering a concept known as an SIDH-Square, then give a brief overview of known attacks on SIDH. This primarily forms and explains the basis for POKÉ. We then cover Kani’s Lemma, a result that is used in POKÉ’s protocol as a main algorithmic tool.

2.1 SIDH Square

Consider two isogenies $\phi : E_0 \rightarrow E_1$ and $\psi : E_0 \rightarrow E_2$ with the same domain and co-prime degree. We define ψ' as the push-forward of ψ under ϕ , where $\psi' : E_1 \rightarrow E_{12}$ has $\ker(\psi') = \phi(\ker(\psi))$. We define $\phi' : E_2 \rightarrow E_{12}$ in a similar way such that $\ker(\phi') = \psi(\ker(\phi))$. We observe that $\psi' \circ \phi = \phi' \circ \psi$, thus the isogenies form some commutative diagram. Drawing a graph of vertices E_0, E_1, E_2, E_{12} , with ϕ, ψ, ϕ', ψ' as edges, we form what is known as an SIDH Square.

$$\begin{array}{ccc} E_0 & \xrightarrow{\phi} & E_1 \\ \psi \downarrow & & \downarrow \psi' \\ E_2 & \xrightarrow{\phi'} & E_{12} \end{array}$$

The Supersingular Isogeny Diffie–Hellman (SIDH) key exchange builds upon this. Like most isogeny-based cryptosystems, it relies on the problem that given the domain E_0 and the codomain E_1 of some isogeny ϕ , it is hard to recover ϕ . In the key exchange, Alice and Bob generate initial isogenies ϕ and ψ respectively, of the SIDH Square. They then share additional information such that Alice and Bob may compute the push-forward isogenies ϕ' and ψ' respectively, completing the square. Then both individuals can arrive at a known E_{12} , which they use to derive a shared session key. For a more complete description of the scheme, the reader may refer to [12].

In 2016, Galbraith et al [10]. proposed an adaptive attack that recovers a user’s static SIDH private key when provided access to an oracle. This necessitated the use of the Fujisaki–Okamoto transform when SIDH was adapted into SIKE during the submission to the NIST PQC standardisation process.

In 2022, Castryck and Decru [4], Maino and Martindale [15] and Robert [19] found an efficient key recovery attack on SIDH, using Kani’s Reducibility Theorem to recover small portions of the unknown isogeny, which they can then combine to recover the full isogeny. This, and a series of SIDH attacks [4, 16], is possible due to the unknown isogeny’s degree and its images on some large torsion basis being known.

The POKÉ algorithm intends to overcome these weaknesses via deriving isogenies of unknown degree, as well as using scalar multiples of any given torsion basis’ images to deter such attacks.

2.2 Kani’s Lemma

Before we give a brief overview of POKÉ, it is appropriate to mention Kani’s Lemma [13]. We follow the theorem cited in POKÉ’s debut paper.

Theorem 1 (Kani’s Lemma [13]). *Consider the following commutative diagram of isogenies in Figure 1.1, where $\deg(\phi) = \deg(\phi') = d_1$, $\deg(\psi) = \deg(\psi') = d_2$, and d_1, d_2 are co-prime.*

Then, the isogeny

$$\Phi = \begin{pmatrix} \phi_1 & -\hat{\phi}'_2 \\ \phi_2 & \hat{\phi}'_1 \end{pmatrix} : E_0 \times E_{12} \rightarrow E_1 \times E_2$$

is a $(d_1 + d_2)$ -isogeny whose kernel is given by

$$\ker(\Phi) = \{([-d_1]P, \phi'_2 \circ \phi_1(P)) \mid P \in E_0[d_1 + d_2]\}$$

The theorem is later used to evaluate images of any point in the domain of the $q(2^a - q)$ -degree isogeny. We will cover this in greater detail in the subsequent section.

Notation 2 We write $\mathbb{Z}_n$ to denote the ring $\mathbb{Z}/n\mathbb{Z}$ and $\mathbb{Z}_n^\times$ to denote the group of invertible elements in $\mathbb{Z}_n$. The group $SL_2(\mathbb{Z}_n)$ is the special linear group, the group of determinant-one 2×2 matrices with entries in $\mathbb{Z}_n$.

3 POKÉ

In this section, we will give a brief overview of the POKÉ cryptosystem. We will also provide a brief explanation as to why it is secure for both Alice and Bob, provided that both parties do not use a static private key value. We will later show how a static key value by Bob can lead to a total leak of Bob's session keys.

A full analysis of POKÉ, including implementation details and a deeper security analysis, can be found in its original paper [2].

We define a prime $p = 2^a 3^b 5^c f - 1$ for some positive integers a, b, c, f . We also define a supersingular curve E_0 over $\mathbb{F}_{p^2}$. Lastly, we sample bases $\langle P_0, Q_0 \rangle = E_0[2^a]$, $\langle R_0, S_0 \rangle = E_0[3^b]$, $\langle X_0, Y_0 \rangle = E_0[5^c]$

It should be noted that the original paper describes the key exchange in three steps. Namely,

- Alice generates her private-public keypair
- Bob generates his keypair, receives Alice's public key, then derives the shared secret and sends additional data over to Alice
- Alice is only able to derive the shared secret upon receiving said additional data

We instead opt to cover it in four parts for further clarity. We begin by describing how Alice, then Bob generate their keypairs. We then explore how Bob is able to derive the shared secret. We end by demonstrating how Alice is able to derive the shared secret upon receiving additional information from Bob.

The last 2 parts will detail a core element of POKÉ which uses Kani's Lemma, as well as discuss the security of POKÉ against conventional SIDH attacks that leak the secret isogeny.

3.1 Alice's Key Generation

The key exchange begins with Alice generating her public-private keys.

Alice samples a random $q \in \mathbb{Z}_{2^a}$, and derives a isogeny $\phi : E_0 \rightarrow E_A$ of degree $q(2^a - q)$. Alice does this using the `FullRepresentInteger` algorithm [6] and an application of Kani's Lemma.

POKÉ's debut paper features a lengthy computation of the above involving Kummer isogenies and theta structures. We will omit this exposition for brevity, as details of it can be found in [2], notably in the source code implementation provided by the authors.

However, we will cover the general concept for completion.

Alice samples random values

$$\alpha_2, \beta_2 \in \mathbb{Z}_{2^a}^\times, \quad \gamma_3 \in \mathbb{Z}_{3^b}^\times, \quad \delta_5 \in \mathbb{Z}_{5^c}^\times$$

and computes points

$$\begin{aligned} P_A &= [\alpha_2]\phi(P_0), & Q_A &= [\beta_2]\phi(Q_0), \\ R_A &= [\gamma_3]\phi(R_0), & S_A &= [\gamma_3]\phi(S_0), \\ \begin{bmatrix} X_A \\ Y_A \end{bmatrix} &= \delta_5 \begin{bmatrix} \phi(X_0) \\ \phi(Y_0) \end{bmatrix}, \end{aligned}$$

where $(\alpha_2, \beta_2, \gamma_3, \delta_5, q)$ is kept as Alice's private key and $(E_A, P_A, Q_A, R_A, S_A, X_A, Y_A)$ is her public key, which she sends over to Bob.

It should be noted that α_2, β_2 must be different, as otherwise the Weil pairing allows for recovery of ϕ [2].

3.2 Bob's Key Generation

Before Bob receives Alice's public key, Bob computes his keys as follows:

Bob samples random $\omega \in \mathbb{Z}_{2^a}^\times$, $r_3 \in \mathbb{Z}_{3^b}$, and $\mathbf{D}_5 \in \text{SL}_2(\mathbb{Z}_{5^c})$ and computes the isogeny $\psi : E_0 \rightarrow E_B$, with $\ker(\psi) = \langle R_0 + [r_3]S_0 \rangle$. We note that the degree of ψ is 3^b .

Bob computes

$$\begin{aligned} P_B &= [\omega]\psi(P_0) \\ Q_B &= [\omega^{-1}]\psi(Q_0) \\ \begin{bmatrix} X_B \\ Y_B \end{bmatrix} &= \mathbf{D}_5 \cdot \begin{bmatrix} \psi(X_0) \\ \psi(Y_0) \end{bmatrix} \end{aligned}$$

where ω^{-1} is the multiplicative inverse of ω modulo 3^b .

Bob keeps (r_3, ω, D_5) as his private key, and $(E_B, P_B, Q_B, X_B, Y_B)$ is his public key. We note that there is no difference between using ω, ω^{-1} and two different ω_1, ω_2 , as an attacker may use the Weil pairing to shift the problem to over ω, ω^{-1} for some ω value [2].

3.3 Bob's Shared Secret

From Alice's public key, Bob takes R_A, S_A and computes a push-forward isogeny $\psi' : E_A \rightarrow E_{AB}$ with $\ker(\psi') = \langle R_A + [\gamma_3]S_A \rangle$.

Bob computes the shared secret

$$\begin{bmatrix} X_{AB} \\ Y_{AB} \end{bmatrix} = \mathbf{D}_5 \cdot \begin{bmatrix} \psi'(X_A) \\ \psi'(Y_A) \end{bmatrix} = \mathbf{D}_5 \cdot \delta_5 \cdot \begin{bmatrix} \psi' \circ \phi(X_0) \\ \psi' \circ \phi(Y_0) \end{bmatrix}$$

As $\mathbf{D}_5$ is needed, intuitively, only Bob can derive the shared secret as of this stage.

In order for Alice to compute the shared secret, Bob sends $(E_{AB}, P_{AB} = [\omega]\phi'(P_A), Q_{AB} = [\omega^{-1}]\phi'(Q_A))$.

The original protocol involves Bob using the shared secret and a key derivation function to encrypt some message that Bob sends to Alice. Alice would then compute the shared secret and decrypt the message accordingly. We will use this as a basis for an oracle in our attack model, where one may check whether the decrypted message is equal to some expected message data.

3.4 Alice's Shared Secret

From the additional information that Alice receives from Bob, since $P_{AB} = [\omega\alpha_2]\psi' \circ \phi(P_0)$ and $Q_{AB} = [\omega^{-1}\beta_2]\psi' \circ \phi(Q_0)$, Alice derives

$$P'_{AB} = [\omega]\psi' \circ \phi(P_0), \quad Q'_{AB} = [\omega^{-1}]\psi' \circ \phi(Q_0)$$

and applies Kani's lemma to compute a push-forward isogeny $\phi' : E_B \rightarrow E_{AB}$ with $\deg(\phi') = q(2^a - q)$. The isogenies ϕ', ψ' are derived such that they form a commutative isogeny diagram, whereby

$$\phi' \circ \psi = \psi' \circ \phi$$

Alice is now able to derive the shared secret

$$\begin{bmatrix} X_{AB} \\ Y_{AB} \end{bmatrix} = \delta_5 \cdot \begin{bmatrix} \phi'(X_B) \\ \phi'(Y_B) \end{bmatrix} = \mathbf{D}_5 \cdot \delta_5 \cdot \begin{bmatrix} \phi' \circ \psi(X_0) \\ \phi' \circ \psi(Y_0) \end{bmatrix}$$

We note that this process requires knowing $\delta_5, \alpha_2, \beta_2$ and q , all of which only Alice knows.

3.5 Deriving The $q(2^a - q)$ Isogeny

This section elaborates on how the $q(2^a - q)$ -degree isogenies ϕ and ϕ' are computed.

We begin by deriving the isogeny ϕ . The **FullRepresentInteger** algorithm in [6] takes an input $M \in \mathbb{Z}$ such that $M > p$ and returns an endomorphism of degree M . This does not work directly on $M = q(2^a - q) < p = 2^a 3^b 5^c f - 1$ unless we reduce b, c , which reduces the possibilities for Bob's r_3 and the shared key X_{AB}, Y_{AB} . This impairs the bit security for Bob's side of the key exchange (3^b), as well as the torsion basis involved in computing the shared key (5^c).

Rather, we use $M = q(2^a - q)3^b$, where we can select large enough c such that the key exchange remains secure and $M > p$ in an overwhelming number of cases. This gives us some isogeny $\theta : E_0 \rightarrow E_0$ that can be separated into isogenies ϕ_1, ϕ_2, ϕ_3 of degrees $q, 2^a - q, 3^b$ respectively. Let $\phi_1 \circ \phi_2 : E_0 \rightarrow E_A$ for some codomain E_A , then $\phi_3 : E_A \rightarrow E_0$.

We can derive the dual isogeny $\hat{\phi}_3 : E_0 \rightarrow E_A$ by computing its kernel, $\ker(\hat{\theta}) \cap E_0[3^b]$.

To compute $\phi(P)$ where $3^b \nmid |P|$ with $|P|$ denoting the order of P , we can perform $[1/3^b]\hat{\phi}_3(\theta(P))$, since $\hat{\phi}_3 \circ \phi_3$ is equivalent to the multiplication-by- 3^b map. This allows us to compute the images of P_0, Q_0, X_0, Y_0 under ϕ but not R_0, S_0 .

Instead, we apply Kani's Lemma, using kernel $\langle ([-q]P_0, \phi(P_0)), ([-q]Q_0, \phi(Q_0)) \rangle$ to obtain the $(2^a, 2^a)$ -isogeny $\Phi : E_0 \times E_A \rightarrow F \times F'$, with ϕ_1, ϕ_2 from before corresponding to the degree q - and $(2^a - q)$ -isogenies.

To determine $\phi(R) \in E_A$ for some $R \in E_0[n], n \in \mathbb{Z}^+$, we compute

$$\begin{aligned}\Phi(R, \mathcal{O}_{E_A}) &= (R'_F, -) \\ \Phi(\mathcal{O}_{E_0}, U_{E_A}) &= (U'_F, -) \\ \Phi(\mathcal{O}_{E_0}, V_{E_A}) &= (V'_F, -)\end{aligned}$$

where (U_{E_A}, V_{E_A}) is a basis of $E_A[n]$.

We can rewrite $R'_F = [x]U'_F + [y]V'_F$ for some $x, y \in \mathbb{Z}_N$.

Since $U'_F = \hat{\phi}_2(U_{E_A})$, we have

$$\phi(R) = R'_{E_A} = \phi_2([x]U'_F + [y]V'_F) = [2^a - q]([x]U_{E_A} + [y]V_{E_A})$$

from which we can derive U_{E_A}, V_{E_A}, x, y . As we also know q and a , we therefore obtain $\phi(R)$.

Hence, we have successfully derived the image of any point P under the $q(2^a - q)$ -isogeny ϕ .

To derive the images of any point X in the push-forward ϕ' , we note that as $\phi' \circ \psi = \psi' \circ \phi'$, P'_{AB}, Q'_{AB} are the images of P_B, Q_B under ϕ' . Hence, the kernel generated by

$$\langle ([-q]P_B, P'_{AB}), ([-q]Q_B, Q'_{AB}) \rangle$$

provides the counterpart Φ' . We then follow the steps mentioned above to derive the image.

3.6 Isogeny-based Security

The general security of the cryptosystem relies on the hardness of computing ϕ or ϕ' from the isogeny domain, codomain, and additional information sent during the key exchange.

We will first cover why it is difficult to recover the shared secret given only the public values.

It has been shown that an isogeny's domain, codomain, as well as additional information of an isogeny's degree and its images on some torsion basis can lead to a complete recovery [4, 16].

At Alice's end of the computations, as the isogenies computed are of unknown degree, the cryptosystem is likely to resist conventional SIDH attacks such as Castryk-Decru. Moreover, the use of $\alpha_2, \beta_2, \gamma_3, \delta_5$ is such that by scaling the images of the basis under the relevant isogenies, it prevents an attacker from

knowing the images of a given torsion basis. This enables Alice's end to resist conventional SIDH attacks as described above. The values γ_3 , δ_5 also ensure that the isogeny degree will not be revealed via pairing computations.

The same can be said for Bob, where the use of a matrix $\mathbf{D}_5$ in place of a scalar adds to the security of the cryptosystem. The scalar ω performs the equivalent of distorting the image of the torsion basis to resist conventional SIDH attacks.

However, the known degrees of ψ and ψ' (3^b) give some useful information. An attacker may be able to, over multiple attempts to conduct a key exchange, piece together Bob's private r_3 value provided it remains constant. They may do so by first deducing r_3 modulo 3, then modulo 3^2 , modulo 3^3 , and so on. We will explore and justify this attack in §4.

4 Adaptive Attack on POKÉ

In this section, we will demonstrate an adaptive attack on POKÉ under the assumption that Bob is using a static r_3 . Note that the authors of POKÉ did not intend for users to set r_3 to be static. We will first show how a dishonest user playing the role of Alice is able to recover Bob's secret and static r_3 through a series of maliciously formed messages simulating a normal transcript in §4.1. Our discussion will cover Bob's key and the points in $E[3^b]$, but the method can be extended to points in $E[m^n]$ for any small prime m and natural number n . We will also show in §4.2 how an adversary is able to compromise other session keys with other users once the adversary is able to recover Bob's static r_3 . In this setting, we will show that generating new secrets ω and $\mathbf{D}_5$ is insufficient to guarantee the security of POKÉ.

We define the following attack model in terms of access to an oracle O , whereby:

- Alice can request that Bob redoes his part of the key exchange with new input values from Alice (under the guise of her repeating her part of the key exchange).
- Bob responds by redoing his part of the key exchange, encrypts some test message m and sends the ciphertext over.
- Alice receives the encrypted test message m , decrypts it on her end, and determines whether the decrypted message and the test message are equal.

We will mount our attack on the above model.

4.1 Recovering r_3

The attacker may begin by recovering Bob's r_3 value using an adaptive attack. They will exploit the following Lemma.

Lemma 3. *Let R and S be linearly independent elements of the group $E[3^n]$ with full order. Then the subgroups*

$$R + [\alpha \cdot 3^{n-1}]S + [r_3]S \quad \text{and} \quad R + [r_3]S$$

are different if and only if $3 \nmid \alpha$.

Proof. As R, S are points of full order and linearly independent, both subgroups have full order equal to 3^b .

If the two subgroups are different, then there must exist an element in one subgroup that does not lie in the other. This implies that the subgroup generators must differ. We also have that

$$R + [\alpha \cdot 3^{n-1}]S + [r_3]S \neq R + [r_3]S$$

which implies that $[\alpha \cdot 3^{n-1}]S \neq 0$. Since the order of S is 3^n , thus $3 \nmid \alpha$.

If $3 \mid \alpha$, for both subgroups to be the same, we require $R + [\alpha \cdot 3^{n-1}]S + [r_3]S = [\lambda]R + [\lambda \cdot r_3]S$ for some $\lambda \in \mathbb{Z}_{3^b}$, but by comparing coefficients and due to the linear independence of R and S , $\lambda = 1$. This means $[\alpha \cdot 3^{n-1}]S = 0$, which implies $3 \mid \alpha$. Contradiction.

Base Step The attacker can apply Lemma 3 to demonstrate recovery of $r_3 \bmod 3$ within 3 oracle accesses.

Alice would normally send R_A, S_A to Bob from which Bob computes $R_A + [r_3]S_A$ as the kernel of his isogeny ψ , the adversary sends $\{R_A - [r \cdot 3^{b-1}]S_A, [1 + 3^{b-1}]S_A\}$ instead for $r = \{0, 1, 2\}$. This causes Bob to compute $R_A + [r_3]S_A + [(r_3 - r) \cdot 3^{b-1}]S_A$ as his kernel generator.

If $r_3 - r \not\equiv 0 \bmod 3$, per Lemma 3, the kernel of ψ' would be different from the kernel generated by $R_A + [r_3]S_A$. Therefore, ψ' will not be the correct push forward of ψ , and Bob's and the attacker's derived shared keys will be different. As a result, when the attacker decrypts the ciphertext, they will obtain a message different from the test message.

Only when $r_3 - r \equiv 0 \bmod 3$ will the kernel subgroup be $R_A + [r_3]S_A$. The key exchange will occur as normal, and the attacker would decrypt and obtain the correct plaintext. This allows us to recover $r_3 \bmod 3$.

Inductive Step Suppose a malicious actor knows the private value $r_3 \bmod 3^k$ for some $k \in \mathbb{Z}^+$. The actor can then recover $r_3 \bmod 3^{k+1}$ from the oracle stated above. The attacker does this with induction.

Let $r_3 \equiv \tau \bmod 3^k$ for some k . Then $r_3 = 3^{k+1} \cdot \lambda + 3^k \cdot r + \tau$ for some $r \in \{0, 1, 2\}$ and $\lambda \in \mathbb{Z}^+$.

An attacker can send

$$\begin{aligned} R'_A &= R_A + [-r' \cdot 3^{b-1} - \tau \cdot 3^{b-1-k}]S_A \\ S'_A &= [1 + 3^{b-1-k}]S_A \end{aligned}$$

in place of R_A, S_A .

An unsuspecting Bob would derive ψ' with kernel

$$\begin{aligned} &\langle R_A + [r_3]S_A + [3^{b-1-k}(r_3 - \tau) - 3^{b-1}r']S_A \rangle \\ &= \langle R_A + [r_3]S_A + [3^{b-1-k}(3^{k+1} \cdot \lambda + 3^k \cdot r) - 3^{b-1}r']S_A \rangle \\ &= \langle R_A + [r_3]S_A + [3^{b-1}(r - r')]S_A \rangle \end{aligned}$$

The last equality holds since the attacker knows the order of S_A is 3^b . From Lemma 3, the attacker knows that this is equivalent to $\langle R_A + [r_3]S_A \rangle$ only when $r \equiv r' \bmod 3$.

The attacker can thus recover r in at most 2 oracle queries. This allows one to gain knowledge of $r_3 \bmod 3^{k+1}$.

By repeating the inductive step b times, a malicious Alice can recover Bob's private r_3 value within $2 \cdot b$ oracle queries.

Remark 4. While the above steps operate on subgroups of order 3^b , we note that they apply to any subgroup of order ℓ^e for some prime ℓ and exponent e . We simply replace 3 with ℓ , b with e and carry on accordingly. The attack will then work in $(\ell - 1) \cdot e$ steps instead of $2 \cdot b$ steps.

4.2 Uncovering new Session Keys

Suppose some other individual, Charlie, performs the same key exchange with Bob, using his own private values $(\alpha_c, \beta_c, \gamma_c, \delta_c, q_c)$ and public key $(E_C, P_C, Q_C, R_C, S_C, X_C, Y_C)$. Let the new basis of $2^a, 3^b, 5^c$ -torsion that Charlie and Bob agree on be $(P_{0C}, Q_{0C}, R_{0C}, S_{0C}, X_{0C}, Y_{0C})$ respectively.

We will show how Alice can recover the new session key generated by Bob with respect to Charlie, after recovering r_3 from Section 4.1.

We begin by tackling a harder variant in which Bob regenerates new random $\omega_C, \mathbf{D}_{5C}$ in his key exchange with Charlie. We note that even if Bob uses the same $\omega, \mathbf{D}_5$ parameters in his exchange with Alice, it does not prevent the attack.

Let $(r_3, \omega_C, \mathbf{D}_{5C})$ be Bob's new private key with Charlie, using different $\omega_C, \mathbf{D}_{5C}$ values. We note that only Bob's isogeny ψ 's codomain E_B is the same. We will thus denote his new public key (relative to Charlie) as $(E_B, P_{BC}, Q_{BC}, X_{BC}, Y_{BC})$.

From r_3 that a malicious Alice has recovered in 4.1, Alice can construct both Bob's isogeny ψ and the push-forward isogeny ψ'_C from the kernel generated using $R_C + [r_3]S_C$.

Since

$$\begin{bmatrix} X_{BC} \\ Y_{BC} \end{bmatrix} = \mathbf{D}_{5C} \cdot \begin{bmatrix} \psi(X_{0C}) \\ \psi(Y_{0C}) \end{bmatrix}$$

Alice can solve for $\mathbf{D}_{5C}$ by performing a change of basis for X_{BC}, Y_{BC} in terms of $\psi(X_C), \psi(Y_C)$. Alice can thus obtain the shared secret

$$\begin{bmatrix} X_{BCC} \\ Y_{BCC} \end{bmatrix} = \mathbf{D}_{5C} \cdot \begin{bmatrix} \psi'(X_C) \\ \psi'(Y_C) \end{bmatrix}$$

and use it to decrypt any traffic between the two.

As for the scenario where Bob's private key uses the same $\omega, \mathbf{D}_5$ as Alice's, Alice can recover $\mathbf{D}_5$ and hence the session key using the same steps above.

4.3 Proof of Concept

A proof of concept was built based on the paper's accompanying code implementation. The current repository of POKÉ as implemented by the authors is found [here](#)

We set up 3 users using `keygenA()`, `encrypt()`, `decrypt()` functions provided in the paper. A malicious Alice and Charlie perform the role of the key exchange's initiator, and Bob plays the role of generating, deriving the shared secret first and relaying information over for Alice/Charlie to derive the shared secret.

For the assumptions in the attack to hold, we make a modification to `encrypt()` allowing for reuse of Bob's private key r_3 . Alice/Charlie may query an oracle, which performs Bob's end of calling `encrypt()` on Alice/Charlie's public keys, which they control. On their own end, Alice/Charlie may call `decrypt()` on the data sent by Bob with their own private keys, and per our assumptions they are able to determine whether or not the shared key they've derived is equal to Bob's.

In the attack, a malicious Alice chooses maligned points and queries the oracle repeatedly till she recovers Bob's r_3 value. Following which, we observe a key exchange protocol done between Bob and Charlie. Alice sniffs the data from the communication and derives the session key between Bob and Charlie. We prove that Alice has derived the correct session key by having Bob encrypt some message to Charlie, which Alice decrypts successfully.

The code for the adaptive attack can be found [here](#).

4.4 Alternate Attack

We propose an alternative, theoretical attack that could work in the same oracle model.

This stems from Theorem 4.1 in [10], which states that an efficient algorithm to solve the isogeny finding problem in an SIDH instance exists if the relevant endomorphism ring is known. This attack idea is also provided courtesy of the authors of [2], who had proposed it to us when we informed them of this paper.

In POKÉ, when Bob uses the same r_3 , we note that the isogeny $\psi : E_0 \rightarrow E_B$ remains the same regardless of Alice's isogeny. Thus, an attacker playing Alice's role in the key exchange may pick two secret isogenies $\phi_0 : E_0 \rightarrow E_{A_0}$ and $\phi_1 : E_0 \rightarrow E_{A_1}$ such that there exists some short isogeny $\rho : E_{A_0} \rightarrow E_{A_1}$ verifying $\rho \circ \phi_0(R_0, S_0) = \phi_1(R_0, S_0)$. Then the curves E_{A_0B} and E_{A_1B} are connected by some short isogeny (the push-forward of ρ under ψ') which can even be brute-forced.

This reveals a non-scalar endomorphism on E_B . By repeating this enough times, the attacker can compute enough information to recover the endomorphism ring of E_B . With a known endomorphism ring of Bob's SIDH instance, we may use the theorem in [10] to recover the isogeny ψ , and by extension, r_3 .

For this attack to be feasible in practice, the endomorphism ring of E_B must be representable efficiently [20].

5 Countermeasures

The first and most obvious countermeasure is for implementations of POKÉ to not allow static private keys to be used. Instead, private keys are to be fully regenerated with every key exchange.

Additionally, an alternative that would allow Bob to maintain and use the same $(r_3, \omega, \mathbf{D}_5)$ private key for all key exchanges would be to adopt an IND-CPA to IND-CCA transform on his data before sending it over to Alice. Ensuring IND-CCA security can be done by applying the Fujisaki-Okamoto transform [8, 11] (as was mentioned in POKÉ [2]), Optimal Asymmetric Encryption Padding (OAEP) [9] (though it must be adapted to fit POKÉ) or similar.

6 Conclusion

In summary, we have demonstrated a possible adaptive attack on POKÉ using a static key. This builds upon the attack mentioned in [10] where a static private key in SIDH (or an SIDH-related protocol) can lead to key leakage. This highlights the significance of CCA security in public-key encryption and key encapsulation mechanisms intended for use in real-world systems.

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